

Effects of plant age on anisotropic saturated hydraulic conductivity of a synthesized rhizosphere using a cubic triaxial permeameter

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Abstract

Understanding the effect of vegetation on anisotropic saturated hydraulic conductivity is essential in the design and performance assessment of bioengineered geostructures. Therefore, this study examines how the age and root biomass of Vetiver grass influence the anisotropic saturated hydraulic conductivity of rhizosphere. Because conventional permeameters cannot reliably measure horizontal and vertical saturated hydraulic conductivity in intact vegetated soils, a recently developed cubic triaxial permeameter was utilized. A total of 76 saturated hydraulic conductivity tests were conducted on specimens cultivated and grown under controlled conditions for 2, 5, and 8 months. The results revealed that increasing plant age and root biomass reduced horizontal saturated hydraulic conductivity by up to 11-fold and vertical saturated hydraulic conductivity by up to two-fold compared with the bare soil, due to root-induced clogging of preferential flow paths. Specifically, the development of dense secondary roots clogs lateral pore spaces, while vertically grown primary roots induce partial preferential flow paths. Although the bare soil exhibits a clear anisotropic behavior ($k_h/k_v > 3$ for void ratio of 0.70), vegetated soils with higher root biomass exhibit almost an isotropic behavior. These findings provide valuable insight for designing and optimizing vegetation-based water management, green landfill covers, and slope stabilization systems.

Key words: saturated hydraulic conductivity, vegetation, anisotropy, cube triaxial permeameter, root biomass

Résumé

Comprendre l'effet de la végétation sur la conductivité hydraulique saturée anisotrope est essentiel pour la conception et l'évaluation des performances des géostructures issues de la bio-ingénierie. Cette étude examine l'influence de l'âge et de la biomasse racinaire du vétiver sur la conductivité hydraulique saturée anisotrope de la rhizosphère. Comme les perméamètres conventionnels ne peuvent pas mesurer de manière fiable la conductivité hydraulique saturée horizontale et verticale dans les sols végétalisés intacts, un perméamètre triaxial cubique récemment mis au point a été utilisé. Au total, 76 tests de conductivité hydraulique saturée ont été réalisés sur des échantillons cultivés dans des conditions contrôlées pendant 2, 5 et 8 mois. Les résultats ont révélé que l'augmentation de l'âge des plantes et de la biomasse racinaire réduisait la conductivité hydraulique saturée horizontale jusqu'à 11 fois et la conductivité hydraulique saturée verticale jusqu'à deux fois par rapport au sol nu, en raison de l'obstruction des voies d'écoulement préférentielles par les racines. Plus précisément, le développement de racines secondaires denses obstrue les espaces poreux latéraux, tandis que les racines primaires verticales induisent des voies d'écoulement préférentielles partielles. Bien que le sol nu présente un comportement anisotrope marqué ($k_h/k_v > 3$ pour un taux de vide de 0,70), les sols végétalisés avec une biomasse racinaire plus élevée présentent un comportement presque isotrope. Ces résultats fournissent des indications précieuses pour la conception et l'optimisation de la gestion de l'eau basée sur la végétation, les couvertures vertes des décharges et les systèmes de stabilisation des pentes.

Mots-clés : conductivité hydraulique saturée, végétation, anisotropie, perméamètre triaxial cubique, biomasse racinaire

1. Introduction

The exploration of soil saturated hydraulic conductivity and its relationships with vegetation has become a principal focus in geotechnical and environmental engineering, particularly given the growing emphasis on sustainable, green, and

environmentally considerate engineering solutions (Bordoloi et al. 2020; Zarei et al. 2024). The rhizosphere, the surficial soil zone influenced by plant roots, plays a critical role in this context as it governs hydrological and mechanical interactions between vegetation and soil, with direct implica-

tions for slope stability, erosion control, and landfill cover performance (Li et al. 2020; Alipanahi et al. 2025b). Vegetation evolves the saturated hydraulic conductivity and reinforces slopes both mechanically and hydrologically through the plant roots (Lann et al. 2024). It has been demonstrated that plant roots contribute to soil cohesion by increasing additional shear strength (Ni et al. 2019). Vegetation is commonly used in geotechnical and geoenvironmental structures such as embankments and landfill cover to prevent shallow slides, limit erosion, and reduce percolation, as seen from recent experimental and reliability-based studies (Xiao et al. 2023; Liu et al. 2024; Qu et al. 2025).

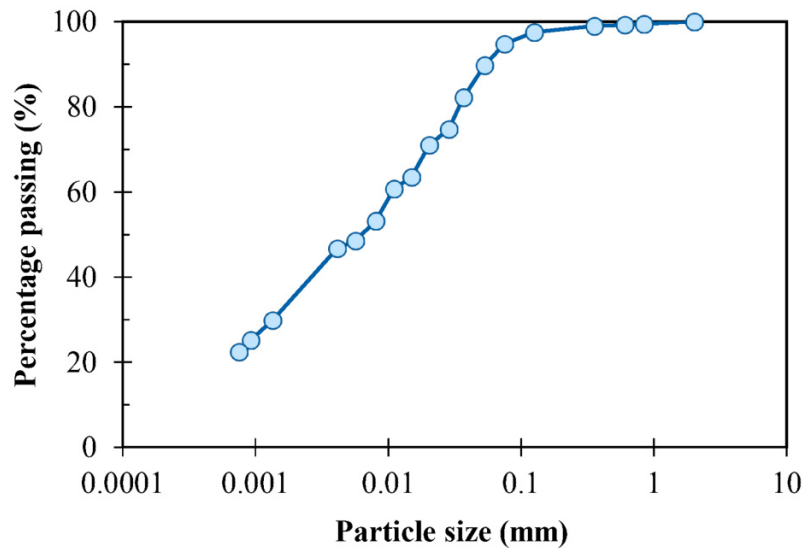
Vegetation significantly enhances the safety factor of slopes, both in terms of mechanical and hydrological effects (Sadeghi et al. 2023; Ng et al. 2024; Yazdani et al. 2024b). The mechanical properties of soil-root composites, often referred to as root cohesion, play a key role in reducing the potential for shallow slope instability (Ng et al. 2015). The magnitude of root cohesion is primarily controlled by plant biomass, which is influenced by factors such as plant age, density, and spacing (Ng et al. 2016a; Ni et al. 2017; Cheng et al. 2023). In addition to mechanical stabilization by root cohesion, vegetation also has a hydrological reinforcement effect by increasing matric suction and changing hydraulic conductivity to improve slope stability and hydraulic performance simultaneously. Through transpiration, vegetation reduces soil moisture and pore water pressure, resulting in an increase in soil shear strength hydrologically (Chen et al. 2019; Ng et al. 2020b; Jabbarzadeh et al. 2024). Vegetation not only affects soil hydrology through transpiration but also influences soil hydraulic properties, such as saturated hydraulic conductivity (Lu et al. 2020). It was reported by Guo et al. (2024b) that the plant reduced rainfall infiltration and preserved slope stability under extreme rainfall for a short period of time. During prolonged rains, the hydrological efficiency of the plant may not be sufficient to maintain slope stability, but the mechanical reinforcement of the roots may be effective. Recent reliability-based research has shown that slope stability is greatly affected by spatial variation in root-induced saturated hydraulic conductivity, and neglecting this variation can lead to unsafe design predictions (Guo et al. 2024a). Additionally, several factors affect the hydraulic properties of vegetated soil, including plant age, root architecture, and plant species (Boldrin et al. 2017; Wang et al. 2020; Zhu et al. 2023). Complementary long-term field observations showed that tree canopies and root systems together regulate soil hydraulic properties and permeability seasonally (Gao et al. 2025). However, the majority of the previous research quantified vegetation effects under simplified or one-dimensional flow conditions, which limits the understanding of root architecture's function in anisotropic saturated hydraulic conductivity.

Saturated hydraulic conductivity plays a crucial role in the prediction of slope failure mechanisms, particularly when soil-plant interactions are taken into account in shallow soils (Darzi et al. 2023; Tsiampousi 2024; Yazdani et al. 2024a). The roots affect soil pore size distribution through three different mechanisms: fissure generation, void clogging, and soil aggregation due to root-chemical interactions (Fraccica et al.

2024). Root penetration, soil particle reorientation, and the formation of macropores as a result of root growth and decay are some of the mechanisms by which vegetation impacts soil microstructure and saturated hydraulic conductivity (Shao et al. 2017). Effects of roots on soil saturated hydraulic conductivity have often shown contrasting results. Plants with lower ages tend to infiltrate water less because roots occupy soil pore spaces, which obstruct water flow paths (Rahardjo et al. 2014; Scholl et al. 2014; Leung et al. 2015; Ng et al. 2016b; Jotisankasa and Sirirattanachai 2017). As well, other research reported higher saturated hydraulic conductivity and infiltration rates when plants are older due to macropores formed by decaying roots or desiccation cracks in high-plasticity clay, which facilitate water flow (Ghestem et al. 2011; Song et al. 2017; Ng et al. 2017b). Recent studies have demonstrated that vegetation alters the pore network and hydraulic characteristics of soil at various scales. Root development was investigated at the micro-scale by X-ray microtomography to modify particle contact geometry and pore connectivity (Wang et al. 2024). In addition, Ng et al. (2014) found no significant differences in saturated hydraulic conductivity between vegetated and non-vegetated soils. A study by Ng et al. (2020a) has shown that saturated hydraulic conductivity is reduced during the first 2–3 months, whereas saturated hydraulic conductivity begins to increase at older plant life stages (13 months). Recent micro- and field-scale observations have provided valuable data on pore geometry and connectivity alteration induced by roots, yet the link between these pore-scale changes and macroscopic anisotropic hydraulic behavior remains under-addressed.

The density of root systems can significantly impact soil saturated hydraulic conductivity. Generally, coarse root systems increase macroporosity, while dense fine root systems increase micropore volume (Bodner et al. 2014). In addition, fine roots may be able to permeate and decrease the volume of large micropores and fine macropores (Scholl et al. 2014). There may have been a blockage of soil pores and a reduction in infiltration due to the small grass roots, whereas larger grass and willow roots may have generated preferential pathways for water to pass through the soil (Leung et al. 2018). Therefore, the composition and density of plant root systems play a crucial role in soil hydrology and are capable of influencing subsurface water flow patterns.

Compacted clays exhibit fabric anisotropy as a result of the majority of flaky clay particles being oriented horizontally (Karimzadeh et al. 2022). As a result, saturated hydraulic conductivity is generally considered to be higher in the horizontal direction than in the vertical direction (Sadeghi et al. 2024a). In most engineered structures, the flow condition is two-dimensional, meaning that information regarding saturated hydraulic conductivity in the horizontal and vertical directions is required. The anisotropic conductivity of vegetated soils plays an important role in studying the hydrological behavior of vegetated slopes. Anisotropic saturated hydraulic conductivity is an essential parameter for reliable analysis of two-dimensional subsurface flows, governing several geoenvironmental problems such as natural slopes, green landfill covers, and vegetated, engineered embankments (Feng et al. 2019; Yildiz et al. 2020; Sitarenios et al.

Fig. 1. Grain size distribution of the soil tested.

2021). It is therefore important to provide some new insights into the characteristics of the anisotropic saturated hydraulic conductivity of vegetated soils.

There are two laboratory methods to evaluate the anisotropic saturated hydraulic conductivity of soils. First, specimens are taken from vertically compacted soil samples in orthogonal directions (Shafiee 2008; Dos Santos and Esquivel 2018), and as a result of this method, specimens are subjected to disturbance, particularly in vegetated soils (Fener and Yesiller 2013). Second, soil samples may be prepared or taken in a cubic mold and tested in a rigid wall permeameter in different directions. However, in conventional rigid wall cells, preferential flow paths, and incomplete saturation have been identified as inevitable sources of error (Namadi et al. 2020; Jia et al. 2022). Despite these advances, no research has critically evaluated the anisotropy of saturated hydraulic conductivity in plant-grown soil, including the influence of plant age and root biomass on it. The lack of test data in multiple directions has caused uncertainties in predicting the behavior of water flow in bio-reinforced soils. A cubic permeameter with a flexible wall should be used to test samples with a sensitive structure to disturbance, such as problematic or vegetated soils (Sadeghi et al. 2024b).

According to the literature surveyed, no systematic study has been conducted to reveal the evolution of anisotropic saturated hydraulic conductivity of the rhizosphere compared with bare soil for a wide range of void ratios. This study is based on the hypothesis that plant growth changes the anisotropy of saturated hydraulic conductivity of soil through root growth and pore structure modification, which can be conducted and explored quantitative using the recently developed cubic triaxial permeameter. Building upon earlier work that established the relationships between vegetation and soil anisotropic saturated hydraulic conductivity under identical testing conditions (Gholami et al. 2025), the present study extends the investigation by focusing on the role of the plant age. In particular, it addresses a key limita-

tion of previous research, which did not account for the temporal evolution of root-soil interactions. Therefore, the purpose of this study is to experimentally investigate the effects of age and root biomass of vetiver grass on anisotropic saturated hydraulic conductivity. Given the intricate nature and variability of root-soil interactions, traditional methods for studying soil saturated hydraulic conductivity are often insufficient, especially in vegetated soils. Hence, a recently developed innovative triaxial permeameter with cube-embedded specimens was used in this study to minimize both natural variability and disturbance to the sensitive root-soil composite structure.

2. Materials and methods

2.1. Soil and plant properties

The soil used in this study was sampled from Tehran, Iran, and was sieved to a size of 2 mm. This soil has a specific gravity of 2.717 and a liquid limit, plastic limit, and plasticity index of 30.8%, 19.6%, and 11.2%, respectively (Ozturk et al. 2022). According to the Unified Soil Classification System, ASTM D2487, this soil is classified as lean clay soil (CL) (ASTM 2009). In addition, according to the Textural Classification (USDA), the soil is classified as silty clay loam. The grain size distribution of the soil used in this study is shown in Fig. 1. Based on the standard Proctor effort (ASTM D698), the maximum dry unit weight and the optimum water content were determined as 17.97 kN/m³ and 16.8%, respectively. For the purpose of determining the essential elements for the growth of plants, Table 1 summarizes the percentage values of major and trace elements constituting the soil.

In this study, Vetiver grass (*Chrysopogon zizanioides* L.) was selected for cultivation in tailor-made cubic molds and testing in a triaxial permeameter. The mold was specifically designed to mimic the cultivation process of the natural rhizosphere under controlled laboratory conditions. The vetiver grass is

Table 1. Weight percentage of vital element for growth condition.

Element	Value (%)
Phosphorus, P	0.05
Potassium, K	0.29
Calcium, Ca	7.55
Magnesium, Mg	1.52
Boron, B	0.01
Zinc, Zn	0.01
Iron, Fe	3.03
Copper, Cu	0.01
Manganese, Mn	0.06

widely found in tropical and subtropical regions, such as India, South, and Southeast Asia. It has an exceptional ability to adapt to harsh environmental conditions, forming deep and extensive roots (up to 5 m depth) within 12 months of growth (Mickovski and van Beek 2009). In addition, Vetiver grass contributes to soil and water conservation and provides a sustainable alternative for ecological restoration (Karimzadeh et al. 2022). These properties make this grass an effective tool for controlling soil erosion, stabilizing slopes, and preventing landslides (Wu et al. 2021).

2.2. Sample preparation and plant cultivation

To prepare cube soil specimens, first, water was sprayed over oven-dried soil to reach a moisture content of 16.8%, and a soil water mixture was left for 24 h for moisture equalization in Ziploc plastic (Motameni et al. 2025; Önal et al. 2025; Alipanahi et al. 2025a). According to Ladd (1978), the prepared mixture was then statically compacted in five layers in a detachable quintuplet-furcation mold with cubic dimensions of 118 mm × 118 mm × 118 mm (Fig. 2a). Approximately 85% relative compaction was adopted and achieved by compacting the soil to a dry density of 1557 kg/m³, which balanced the conditions for root growth and shear strength requirements according to Ng et al. (2014, 2017a). Through cross-scarification of layers, the adverse effects of layering on saturated hydraulic conductivity measurements have been reduced.

The vegetated specimens (G) were prepared by digging a hole 30 mm in diameter and 30 mm deep, planting Vetiver grass, and carefully pouring the retrieved soil into the hole. Most of the soil that had been excavated was filled in to maintain the root zone and the surrounding matrix at consistent dry density, with a small portion excluded to account for the space occupied by the roots. Although some slight disturbance is unavoidable during transplanting, the same procedure was applied to all specimens to ensure consistency in preparation. Following transplantation, it was necessary to irrigate the plant specimens every day for the first month, and then twice a week thereafter. The irrigation mode was set from previous tests and direct observation to maintain a steady state of low moisture characteristic of early plant growth. To each cycle, approximately 90 mL of water was added to each specimen, or approximately 2.9% water con-

tent and 10% degree of saturation. Bare specimens (B) had been subjected to the same irrigation scenarios as vegetated specimens for the period of 8 months.

The vegetated specimens (G) were grown to the target age of 2, 5, and 8 months (Figs. 2a and 1b) and subsequently were removed from the molds (Figs. 2b and 2c). The grass shoots were carefully cut and removed to create an ideal cube for anisotropic saturated hydraulic conductivity tests (Fig. 2c). The cube design has the merit that tests can be conducted on the vegetated specimens in any desired orthogonal direction, so that anisotropy ratio with respect to conductivity can be determined. There was no fertilizer added during the growth phase to prevent induced osmotic suction caused by solutes in the soil pore water (Yan and Zhang 2015; Ng et al. 2016b).

The vegetated specimens were grown in the cultivation room of the Green Geotechnology Laboratory at Sharif University of Technology. Temperatures and relative humidity were constant at 29 °C and 60%, respectively, throughout the day. A cultivation room was set up to simulate a day for 12 h with cool and warm white fluorescent lighting, and a night for 12 h without lighting. Following saturated hydraulic conductivity tests, vegetated specimens were submerged in water for 1 day, and the grass was separated from the soil by washing the roots with a 2 mm sieve. Accordingly, the roots were initially air-dried and weighed in air, then immersed in water to obtain their apparent weight. The root volume was determined using Archimedes' principle, based on the difference between the weights measured in air and in water.

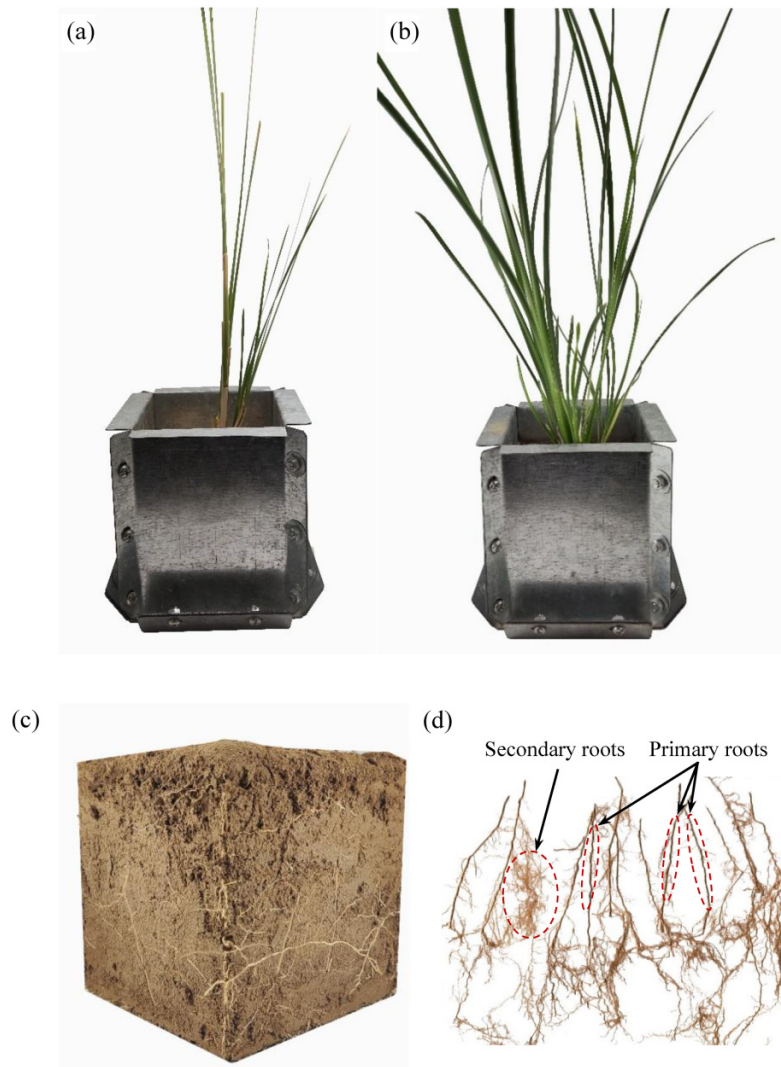
Figure 2d illustrates the root system morphology obtained following the procedure described. There are two types of roots: primary roots and secondary roots. Primary roots have a thickness greater than 1 mm, while secondary roots have a thickness less than 1 mm (Leung et al. 2018). To determine the length of the primary and secondary roots, an image processing software (ImageJ) was used. Based on visual inspection of destructured vegetated specimens after testing, primary roots grow vertically, while secondary roots grow horizontally in the soil matrix, consistent with typical root system architecture patterns reported in the literature (Leitner et al. 2010).

2.3. Experimental procedure

To determine the anisotropic saturated hydraulic conductivity of both bare and vegetated specimens without sampling and potential disturbance, a tailor-made cubic sample embedded triaxial permeameter was used (Arabchobdar et al. 2024; Gholami et al. 2025). The specimen was placed between two adapters, two porous stones, and two filter papers in this experiment. The specimen was then wrapped in a membrane, followed by the connection of the pore-water pressure and the cell pressure lines. The triaxial cell was installed and subsequently filled with de-aired water in the cell. A seating confining stress level, as low as 15 kPa, was applied, and de-aired water was circulated from the bottom to the top of the specimen under a 10 kPa pressure gradient until steady-state flow was achieved (Irani et al. 2024).

An increase in cell and back pressure at intervals of 10 kPa was applied to the specimen prior to the experi-

Fig. 2. Typical vegetated specimens: (a) specimen with low root biomass, (b) specimen with high root biomass, (c) unmolded cubic specimen, and (d) root system morphology of a vegetated specimen.



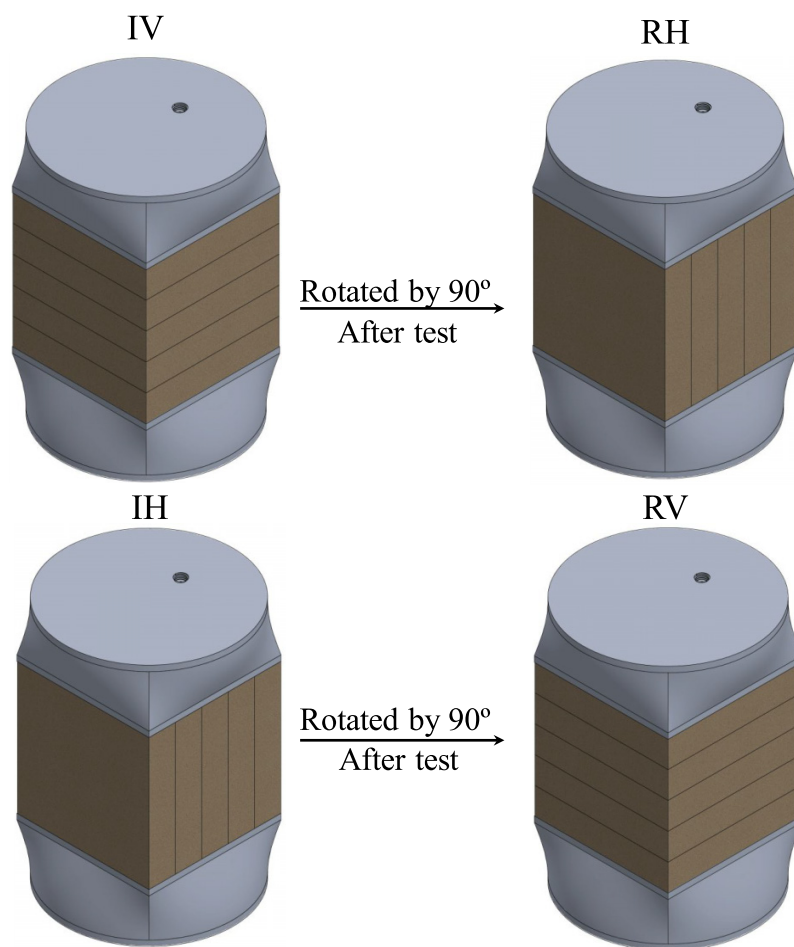
ment to achieve back-pressure saturation (Sadeghi and Alipanahi 2020). According to ASTM D5084, back pressure was increased until a B-value of 0.95 or greater was recorded for three readings. The specimens were consolidated at target confining pressures of 15, 25, 50, 75, and 100 kPa, and volume changes were recorded (Abuhajar et al. 2015). Saturated hydraulic conductivity experiments were conducted using a cubic sample embedded in a triaxial permeameter apparatus according to ASTM D5084. During the saturation procedure, as well as before the measurement of saturated hydraulic conductivity, the volume changes of specimens were measured. All tests were conducted at a temperature of 24 °C.

In the cube permeameter, the specimen can either be seated vertically or horizontally after compaction, i.e., rotated by 90 degrees in comparison to the direction of compaction. Figure 3 illustrates the mechanism of the anisotropic saturated hydraulic conductivity tests and how the four types of tests were conducted either in the direction of compaction or orthogonally. Initial (I) tests are conducted on specimens after they were prepared, such as IV and IH tests. In the sec-

ond experiment, the specimens were disassembled from the permeameter and seated in the rotated (R) direction, including RV and RH. If the specimen were seated vertically first (IV), it would be rotated and seated horizontally (RH), and the second saturated hydraulic conductivity test would be conducted in a horizontal direction to omit the effect of natural variability. In the case of specimens seated horizontally first (IH), they were rotated and seated vertically (RV), and the subsequent saturated hydraulic conductivity tests were conducted in the vertical position. To maintain exactly the same biomass properties of vegetated specimens, anisotropic saturated hydraulic conductivity was measured on identical specimens using this test procedure.

2.4. Test plan

Triaxial saturated hydraulic conductivity tests were conducted to evaluate the anisotropic hydraulic conductivity of bare (B) specimens and to examine the effect of Vetiver roots on vegetated (G) specimens. The tests were conducted on specimens that had been cultivated for 2, 5, and 8 months,

Fig. 3. Schematic overview of specimen positioning and anisotropic hydraulic conductivity testing sequence.

and the root biomass was determined after the test. The test plan was developed to evaluate the influence of plant age on anisotropic saturated hydraulic conductivity while accounting for root development variability. A total of eight specimens were tested—one at 2 months, two at 5 months, three at 8 months, and two bare soil specimens. The number of tested specimens was increased for older plants, because the variability in root biomass and morphology became more pronounced with plant growth and both well-developed and poorly developed roots were observed. Each specimen was tested in both vertical and horizontal orientations in initial (I) and rotated (R) configurations, and each test series included hydraulic conductivity measurements at various void ratios by raising the confining pressures (15, 25, 50, 75, and 100 kPa). In total, 76 measurements were obtained. An overview of the test conditions, as well as the measured root biomass and soil void ratio, is provided in Table 2. The cubic triaxial permeameter enables horizontal and vertically saturated hydraulic conductivity measurements using identical vegetated samples and minimizes the effect of natural variability. This approach allowed reliable replication and direct comparison within the same root structure, as growing identical vegetated samples is highly unlikely, as reflected by the variability shown in Table 3.

3. Interpretation of the results

3.1. Root characteristics

The relationship between plant age and both root biomass and root volume ratio across six specimens with 2, 5, and 8 months is presented in Fig. 4a. Despite being cultivated under consistent environmental conditions, the data reveal significant variability, with no clear or consistent upward trend over time. While some specimens show an increase in root biomass and root volume ratio with age, this pattern is not uniform across all specimens, contrary to expectations of a monotonic increase in root biomass and volume ratio with plant age. This inconsistency could be due to the intrinsic biological variability of plants affecting each specimen. For the measurement of data variability, standard deviation (SD) and coefficient of variation (COV) were calculated for root biomass and root volume ratio at 5 months as well as 8 months of plant age. This variability suggests that plant age alone is not a reliable predictor of root development; instead, biomass and root morphology must be directly measured to capture vegetation effects on soil hydraulic properties accurately. On the other hand, Fig. 4b illustrates a strong linear relationship between root biomass and root volume ratio across all specimens. The data follow a clear trend, indicating that

Table 2. Summary of test program and conditions for triaxial hydraulic conductivity tests.

Test series*	Angle	Age (month)	Hydraulic conductivity measurement stress levels (kPa)	Initial conditions	
				Void ratio	Root biomass (kg/m ³)
B-V	0	8	25, 50, 75, 100	0.77	N.A.
B-H	90	8	15, 25, 50, 75, 100	0.78	N.A.
G-V-1.50-A2	0	2	15, 25, 50, 75, 100	0.79	1.50
G-V-2.50-A5	0	5	25, 50, 75, 100	0.80	2.50
G-H-1.33-A5	90	5	15, 25, 50, 75, 100	0.81	1.33
G-V-0.98-A8	0	8	15, 25, 50, 75, 100	0.79	0.98
G-H-2.32-A8	90	8	15, 25, 50, 75, 100	0.82	2.32
G-H-1.29-A8	90	8	15, 25, 50, 75, 100	0.80	1.29

*B: bare, G: vegetated, H: horizontal, V: vertical, A: age, and N.A.: not applicable

Table 3. Biotic properties of vegetated soils samples.

Test series	Root biomass (kg/m ³)	Root volume ratio (m ³ /m ³)	Primary roots length (m)	Secondary roots length (m)	Total roots length (m)
G-V-1.50-A2	1.503	0.0029	4.33	13.30	17.63
G-V-2.50-A5	2.504	0.0048	6.87	36.72	43.59
G-H-1.33-A5	1.329	0.0025	4.56	8.32	12.88
G-V-0.98-A8	0.977	0.0019	3.96	5.94	9.90
G-H-2.32-A8	2.317	0.0044	6.91	25.87	32.78
G-H-1.29-A8	1.295	0.0025	4.47	7.13	11.60

root density, defined as the ratio of root biomass to root volume, is consistent across specimens, confirming the authentic measurements of weight and volume.

Figure 5 displays the relationship between root biomass and the lengths of primary and secondary roots, as well as the ratio of secondary to primary root lengths across specimens. A positive correlation between root biomass and root length for both primary and secondary roots was indicated, with secondary roots showing a steeper increase in length relative to primary roots. The ratio of secondary to primary root lengths increases with root biomass, confirming that secondary roots grow proportionally longer than primary roots as the root system develops. This trend suggests a shift in root system architecture with increasing biomass, where secondary roots become more dominant. This structural shift in root architecture is consistent with prior findings (Leung et al. 2018) that lateral roots dominate pore clogging and restrict horizontal conductivity. This increasing ratio points to the potential for secondary roots to play a larger role in occupying and clogging soil pore spaces, particularly in the horizontal direction, which could reduce horizontal saturated hydraulic conductivity. A comprehensive critique of the different behaviors of primary and secondary roots was proposed by Gholami et al. (2025). Biotic properties of the sixth vegetated soils with different ages are presented in Table 3 in detail. It should be noted that the present study focused on the active growth and maturity phases of Vetiver within 8 months, during which no root decay was observed after the test as shown in Fig. 2d. While the results are representative of growing vegetation conditions, further investigation is required to assess the long-term effects of root decay and seasonal changes on hydraulic anisotropy and soil structure.

3.2. Effects of plant age on compressibility of specimens

Figures 6a and 6b show the compression curves of bare and vegetated soil specimens, with specimens seated in vertical and horizontal orientations. The compression behavior of bare (B-V and B-H) compared with vegetated specimens at various root biomasses. Gholami et al. (2025) showed that comparable pairs of B-H and B-V follow the same trends ideally under isotropic consolidation, which makes them a reliable cornerstone for studying the vegetated specimens.

In vertically seated specimens, the bare soil (B-V) shows a relatively steady decrease in void ratio as effective stress increases. Vegetated specimens with different root biomasses of 1.50, 2.50, and 0.98 kg/m³ exhibit a suppressed compressibility, but at consistently higher values than the bare soil. This difference, which is a direct result of the varying root biomasses, clearly demonstrates the significant influence of roots on soil compressibility. Similarly, horizontally seated vegetated specimens, with root biomasses of 1.33, 1.29, and 2.32 kg/m³, exhibit similar trends to the vertically oriented specimens, as shown in Fig. 6b. The compression curves indicate that vegetated soils, particularly those with higher root biomass, display reduced compressibility in both vertical and horizontal orientations. The root structure, especially at higher biomasses, evidently reinforces the soil matrix, limiting particle rearrangement, and reducing compressibility in the vertical orientation. Across both orientations, vegetated specimens with higher root biomass (e.g., G-V-2.50-A5 and G-H-2.32-A8) maintain the highest void ratios under stress. This suggests that greater root biomass contributes to increased soil structural strength, providing additional resistance un-

Fig. 4. Biotic characteristics of vegetated soils: (a) effect of plant age on root biomass and root volume ratio, and (b) correlation between root biomass and root volume ratio.

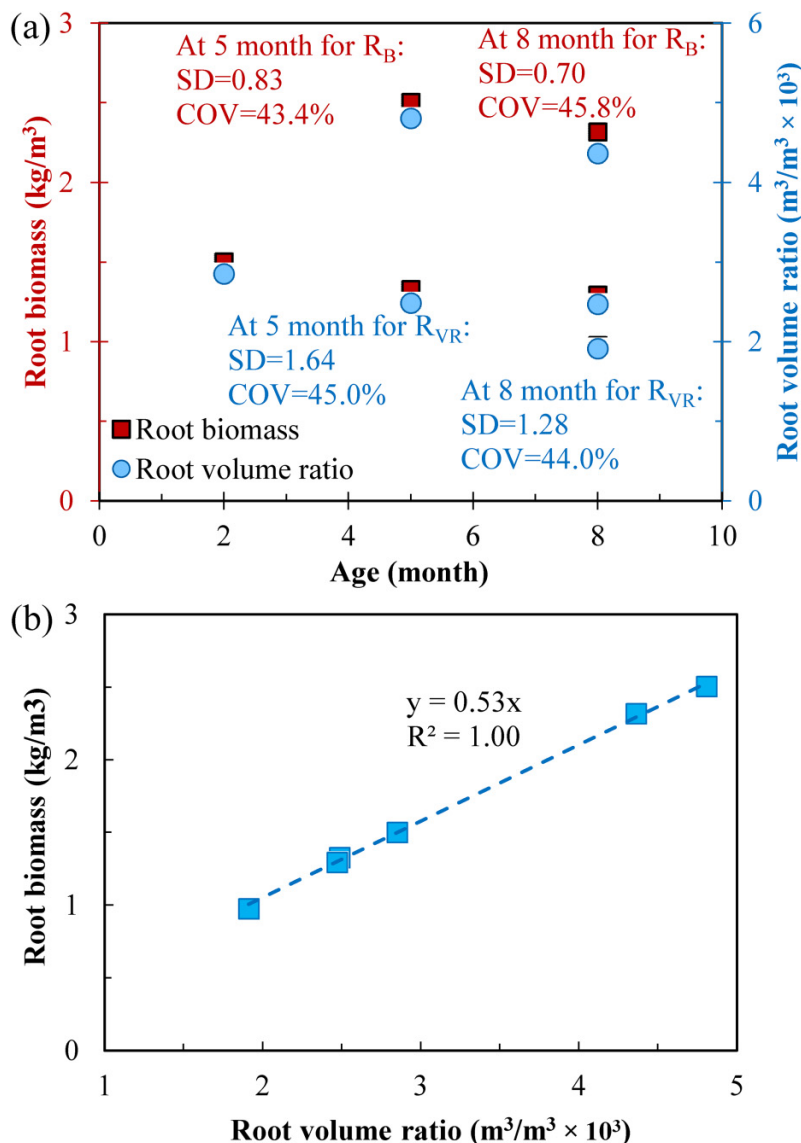


Fig. 5. Correlation between root biomass and the roots length, and secondary-to-primary roots length ratio.

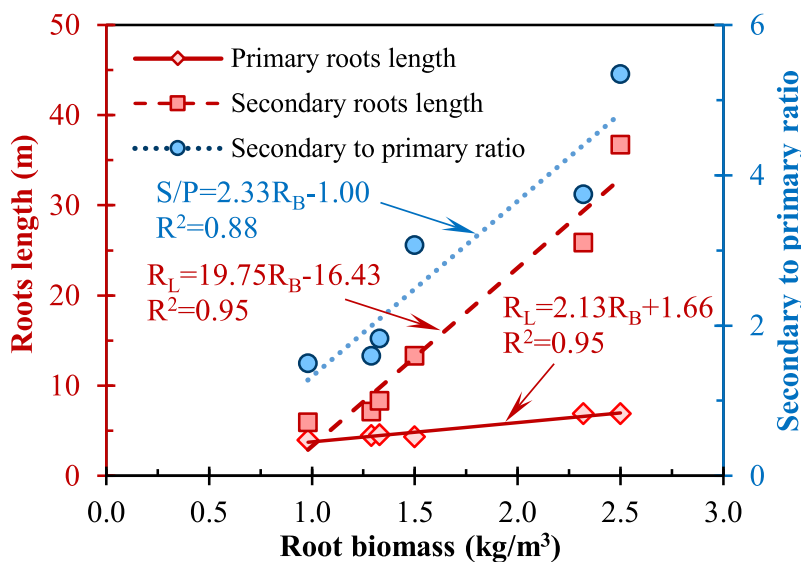
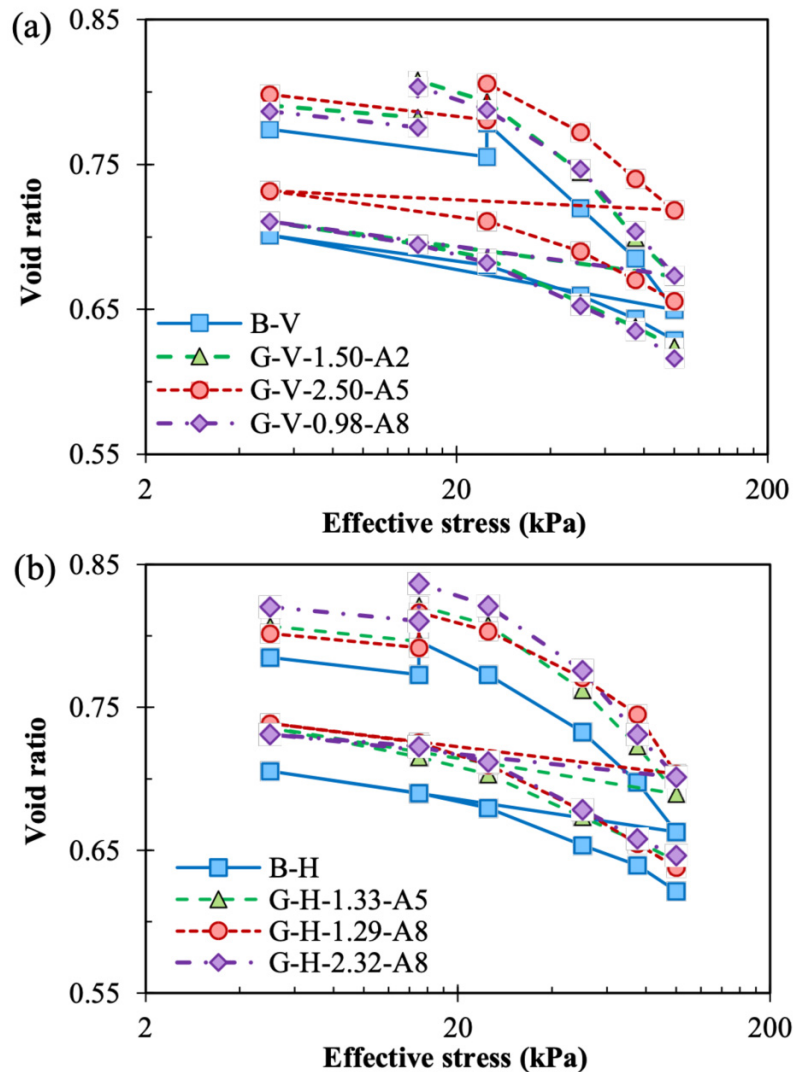


Fig. 6. Compression curves of bare and vegetated specimens seated (a) vertically and (b) horizontally.



der isotropic compression. Similar root-reinforced effects on fabric anisotropy have been reported by Shao et al. (2017).

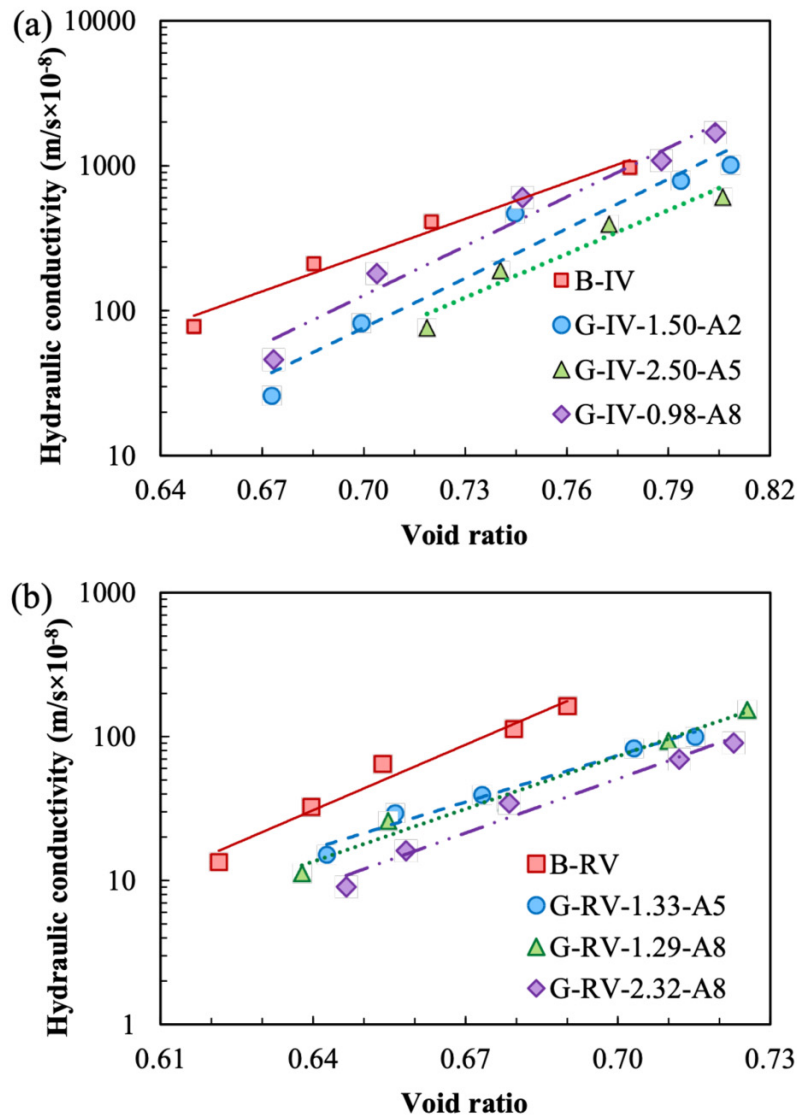
3.3. Effects of plant age on anisotropic saturated hydraulic conductivity

Figures 7a and 7b illustrate the relationship between vertical saturated hydraulic conductivity and void ratio for bare (B-IV and B-RV) and vegetated soil specimens (G-IV and G-RV) across different root biomasses. The measurements were taken with specimens in both initially vertical (IV) and rotated vertical (RV) orientations, providing insights into how root structure affects soil vertical saturated hydraulic conductivity. Across both types of specimens, bare soil specimens (B-IV and B-RV) display higher saturated hydraulic conductivity values than vegetated soils at equivalent void ratios, with vegetation reducing the vertical saturated hydraulic conductivity by up to twofold. This difference indicates that root presence generally reduces vertical saturated hydraulic conductivity by reducing pore spaces and restricting water

flow. The reduction effect is more pronounced in specimens with higher root biomass, as observed in specimens G-IV-2.50-A5 and G-RV-2.32-A8, which exhibit the lowest conductivity values. It should be noted that, in Fig. 7a (IV orientation), vegetated specimens show a moderate decrease in saturated hydraulic conductivity at the high range of void ratio, likely because root structure in the vertical orientation induced some flow channels to create, particularly through primary roots.

Similarly, Figs. 8a and 8b showed the effects of root growth on horizontal saturated hydraulic conductivity, comparing vegetated specimens to bare ones. Across both initially horizontal (IH) and rotated horizontal (RH) orientations, vegetated soils consistently exhibit lower horizontal saturated hydraulic conductivity than bare soils at equivalent void ratios. This reduction is most pronounced in high root density specimens (e.g., G-IH-2.32-A8 and G-RH-2.50-A5), where roots occupy horizontal pore spaces and restrict flow, and decrease horizontal saturated hydraulic conductivity by up to 11 times.

Fig. 7. Vertical hydraulic conductivity of bare, and vegetated specimens: (a) initially vertical and (b) rotated vertical positioned samples.

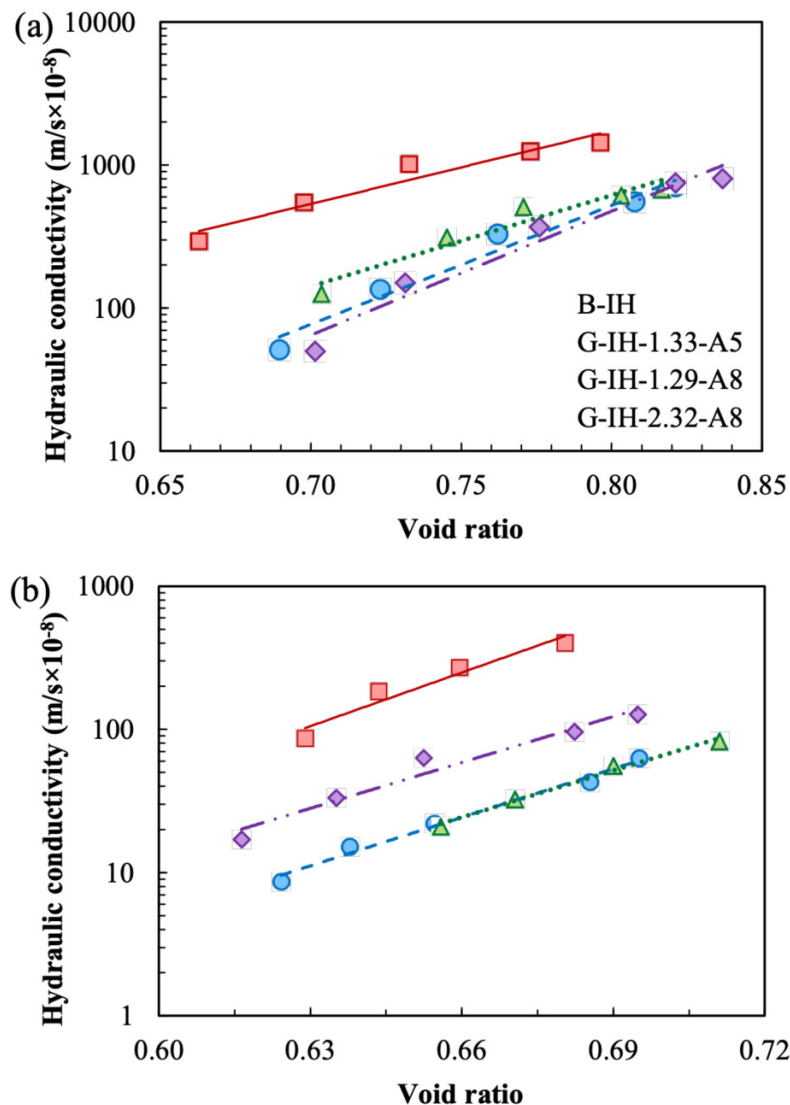


Conductivity in rotated specimens (RV and RH) is lower than in initial specimens (IV and IH), mainly due to the compressed structure resulting from prior isotropic loading and subsequent rotation. This effect is especially notable in higher biomass specimens, where the dense root network occupies the reduced void spaces, further limiting water flow. While all specimens show a decrease in conductivity with enhanced confinement, vegetated soils exhibit a more gradual decrease rate compared to bare soils. This indicates the growth of root networks in vegetated soils buffers the impact of void ratio changes on conductivity, stabilizing hydraulic behavior across varying void ratios.

Saturated hydraulic conductivity functions of bare and vegetated soil specimens plotted against root biomass across a range of void ratios ($e = 0.65, 0.69, 0.73, 0.77$, and 0.81) in vertical and horizontal orientations, as shown in Figs. 9a and 9b. The exponential trend lines for each void ratio capture the relationship between horizontal and vertical saturated hydraulic conductivity and root biomass, high-

lighting how root density impacts soil conductivity at a specific void ratio. Across all void ratios and orientations, saturated hydraulic conductivity decreases exponentially with increasing root biomass. This trend is consistent in both vertical and horizontal orientations, indicating that root biomass has a substantial impact on reducing soil saturated hydraulic conductivity. In both figures, the vertical orientation generally shows higher conductivity than the horizontal orientation for each void ratio and biomass level. This difference may be attributed to the alignment of primary roots in the vertical direction, which provides channels that facilitate some degree of vertical water movement. In contrast, water flow in the horizontal direction is more impeded by lateral root structures, especially at higher root biomass levels, leading to reduced saturated hydraulic conductivity. The horizontal and vertical saturated hydraulic conductivity functions can be used to model the vegetated soils in different void ratios, representing the depth.

Fig. 8. Horizontal hydraulic conductivity of bare, and vegetated specimens: (a) initially horizontal and (b) rotated horizontal positioned samples.



3.4. Evolution of anisotropic saturated hydraulic conductivity ratio

Figure 10 depicts the evolution of the anisotropy ratio (k_h/k_v) for B and G specimens with varying root biomass (0.98, 1.33, 1.50, 2.32, 2.50 kg/m^3) across a range of void ratios. The figure highlights the impact of vegetation on the anisotropy of saturated hydraulic conductivity as the void ratio changes. The results are inferred from the vertical and horizontal saturated hydraulic conductivity functions as depicted in Figs. 7 and 8. In bare soil, the anisotropy ratio is roughly around one at the initial state and increases as the void ratio decreases under isotropic consolidation. At lower void ratios, the anisotropy ratio in bare soil reaches values as high as 5, reflecting significant horizontal flow dominance. It is due to the lower resistance of the degraded zone on the upper part of the specimen, as observed and reported by Gholami et al. (2025).

In vegetated soils, on the other hand, the anisotropy ratio stabilizes near the isotropic reference line across all void

ratios, regardless of root biomass. This demonstrates that vegetation eliminates the preferential horizontal flow observed in bare soil, creating a more isotropic flow behavior. In fact, vegetation prevented the degraded zone created towards the upper part of the soil specimen. Moreover, roots contribute to isotropy by obstructing horizontal flow paths while creating vertical channels by primary roots, regardless of the void ratio or root biomass. This behavior highlights the stabilizing effect of roots on soil hydraulic properties, particularly in reducing directional differences in conductivity.

Figure 11 shows the evolution of the anisotropy ratio as a function of root biomass at a constant void ratio of 0.70. The void ratio of 0.70 was chosen because it falls within the range of all specimens tested, enabling a fair comparison of the anisotropy ratio across different root biomasses under similar compaction and stress conditions. The anisotropy ratio provides insight into the influence of root growth on directional variation in saturated hydraulic conductivity. Under

Fig. 9. Hydraulic conductivity as a function of root biomass for wide range of void ratios along (a) horizontal, and (b) vertical directions.

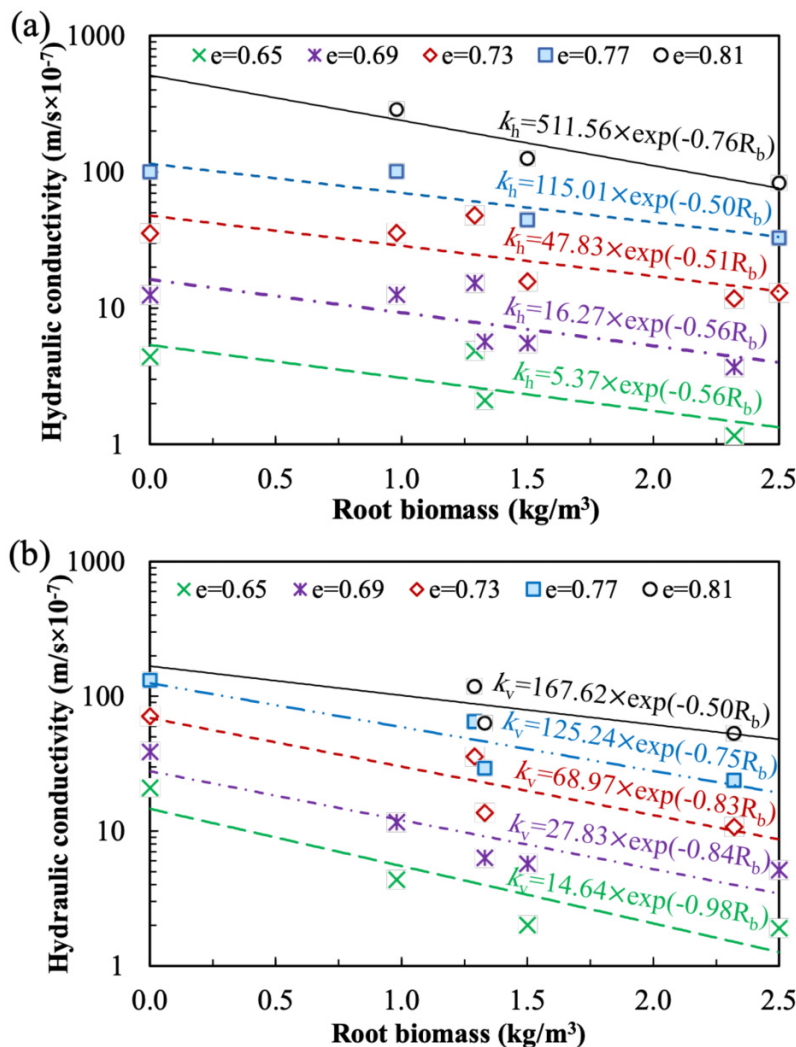


Fig. 10. Evolution of the anisotropy ratio of bare, and vegetated specimens with void ratio.

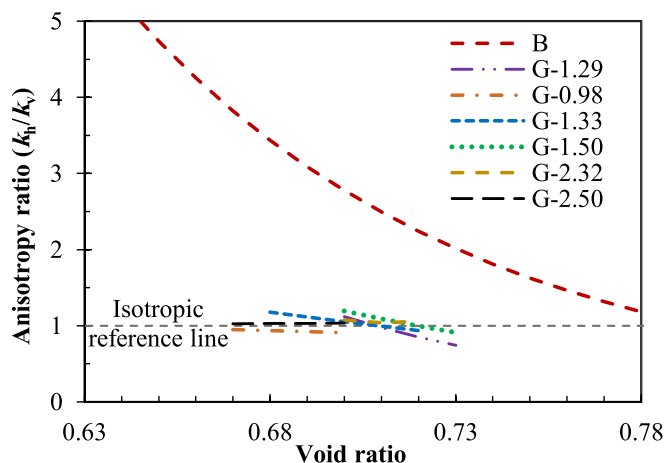


Fig. 11. Evolution of the anisotropy ratio with root biomass for the specific void ratio of 0.70.

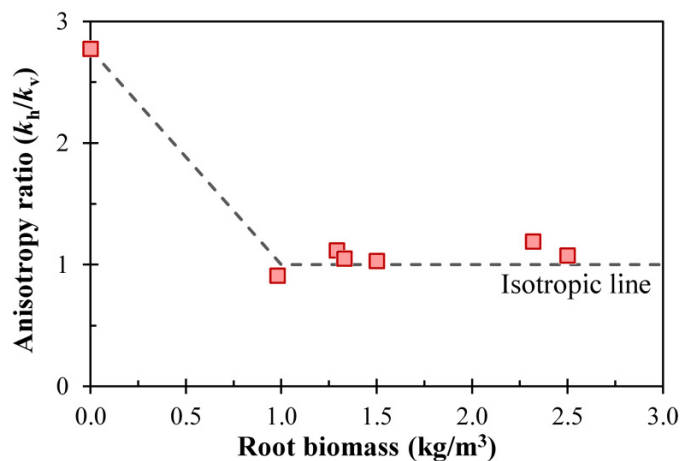
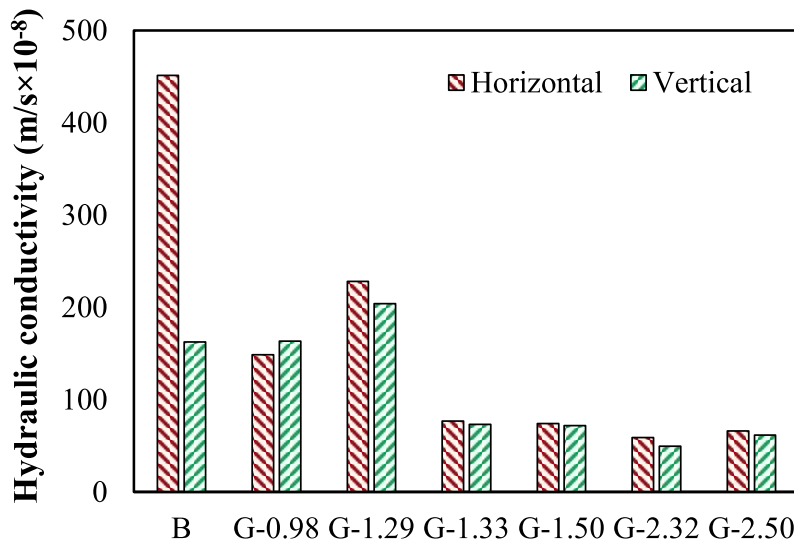


Fig. 12. Comparison of orthogonal hydraulic conductivities of bare, and vegetated specimens for the specific void ratio of 0.70.

bare conditions, the anisotropy ratio is significantly greater than 1 ($k_h/k_v \approx 3$), indicating a strong preference for horizontal conductivity in bare soil. This result is consistent with bare soil behavior, where structural features such as degraded zones or horizontal cracks promote horizontal water flow. As root biomass increases, the anisotropy ratio decreases sharply, approaching the isotropic line ($k_h/k_v = 1$). This shift from anisotropic to isotropic state suggests that roots disrupt the preferential horizontal flow paths, equalizing conductivity in both directions. This observation reveals that root networks lead to balanced conductivity in horizontal and vertical directions, possibly due to the homogenizing effects of root-induced pore clogging. In other words, the roots play a critical role in suppressing anisotropy with respect to saturated hydraulic conductivity by disrupting horizontal flow paths.

Figure 12 presents the horizontal and vertical saturated hydraulic conductivity of B and G specimens at the void ratio of 0.70. The results align with the trends observed in previous figures. The horizontal saturated hydraulic conductivity in bare soil is significantly higher than the vertical conductivity, confirming the strong anisotropy previously observed in Figs. 10 and 11. In vegetated specimens (G-0.98 to G-2.50), both horizontal and vertical hydraulic conductivities are significantly reduced compared to bare soil, with minimal differences between the two directions. Even at lower root biomass (e.g., G-0.98), vegetation effectively balanced horizontal and vertical conductivity. As root biomass increases (e.g., G-2.50), both horizontal and vertical conductivities stabilize at similarly low values, indicating that denser root networks consistently obstruct flow paths in all directions, regardless of orientation. In summary, vegetation transforms the anisotropic behavior of bare soil, designating with a ratio as high as 5, into isotropic behavior by balancing horizontal and vertical hydraulic conductivities. Root networks play a critical role in reducing conductivity and mitigating directional differences, even at lower biomass levels. These findings that vegetation consistently reduces anisotropy across ages and

void ratios challenge the assumption that roots always induce anisotropy (Ghestem et al. 2011; Leung et al. 2015). Accordingly, vegetated slopes, landfill covers, and green embankments can be modeled assuming near-isotropic conductivity once root networks are established. Moreover, the homogenizing effect of vegetation reduces risks of preferential piping or anisotropy-driven instability.

4. Conclusions

The effects of plant age within the active growth stage on the anisotropic saturated hydraulic conductivity of vegetated soils were investigated in this study using a cubic triaxial permeameter, based on a total of 76 saturated hydraulic conductivity tests. All specimens were prepared under identical environmental and compaction conditions, with plant age being the sole varying factor, ranging from 1 to 8 months. The key conclusions drawn from this study are as follows:

1. Vegetation, particularly in older plant specimens, significantly strengthens the composite soil-root matrix against isotropic compression by providing an extra skeleton force chain of the root system. This lessened sensitivity to isotropic loading suggests that root networks act as reinforcing elements within the soil matrix, stabilizing the structure and reducing compressibility.
2. The effect of plant age on the anisotropic saturated hydraulic conductivity of vegetated soils is indirect and is primarily manifested through age-related variations in root development rather than chronological age itself. While no consistent linear correlation was found between plant age and root biomass, specimens from older age groups generally exhibited lower saturated hydraulic conductivity and reduced anisotropy. This behavior can be attributed to the progressive formation of a denser and more interconnected root network with age, which modifies the soil pore architecture and limits preferential flow paths. Therefore, plant age governs the hydraulic behavior.

ior of vegetated soils through its influence on root growth dynamics, specifically, the extent, distribution, and structure of the root system. This distinction highlights the importance of coupling biological maturity indicators with age when evaluating vegetation effects on soil hydraulic properties.

3. Root biomass ranged from 0.98 to 2.50 kg/m³, without a consistent relationship being observed between biomass and age. This reflects the intrinsic natural variability of plant growth under identical conditions. Nonetheless, a strong linear correlation ($R^2 = 1.00$) was found between root biomass and root volume ratio, indicating consistency in root density across specimens.
4. Both vertical and horizontal hydraulic conductivities decreased exponentially with increasing root biomass, demonstrating the strong influence of root density on the diminishing of pore-network connectivity. In rotated specimens, additional reductions in conductivity were observed due to the compression history and lower void ratios. These results highlight that biomass-driven modifications dominate over stress-path effects in governing hydraulic response.
5. Of particular interest, all vegetated specimens showed anisotropy ratios between 0.9 and 1.1, indicating nearly isotropic behavior. In contrast, the bare soil exhibited a maximum anisotropy ratio approaching 5. The results demonstrate that vegetation, regardless of age, is effective in reducing saturated hydraulic conductivity and the anisotropy ratio of the rhizosphere compared with the corresponding compacted bare soil. The homogenizing effect of vegetation arises from lateral roots clogging horizontal pathways and vertical roots forming compensating channels, thus balancing directional conductivities.

Overall, the findings are representative of the active growth and maturation phases of vegetation. Future studies should extend this work to include the decay phase and long-term cyclic effects to extend these findings to post-maturity conditions.

Acknowledgements

This work is based upon research funded by Iran National Science Foundation (INSF) under project No. 4035488.

Article information

History dates

Received: 12 September 2025

Accepted: 26 December 2025

Accepted manuscript online: 7 January 2026

Version of record online: 24 February 2026

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Data availability

All data supporting the study's findings can be accessed from the corresponding author upon reasonable request.

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Competing interests

The authors have read and approved this version of the article, and due care has been taken to ensure the integrity of the work. They also declare that the work described is original research that has not been published previously and is not under consideration for publication elsewhere, in whole or in part.

References

- Abuhajar, O., Newson, T., and El Naggar, H. 2015. Scaled physical and numerical modelling of static soil pressures on box culverts. *Canadian Geotechnical Journal*, **52**(11): 1637–1648. doi:[10.1139/cgj-2014-0493](https://doi.org/10.1139/cgj-2014-0493).
- Alipanahi, P., Mirzaee, P., Mirsadeghi, M.N., Masrou, F.F., Yazdani, F., Zornberg, J.G., and MolaAbasi, H. 2025a. Durability and life-cycle evaluation of high-plasticity clays treated with zeolite as partial replacement of cementitious stabilizers. *Results in Engineering*, **28**: 108155. doi:[10.1016/j.rineng.2025.108155](https://doi.org/10.1016/j.rineng.2025.108155).
- Alipanahi, P., Motameni, S., Yazdani, F., and Zhang, L. 2025b. Numerical analysis of the effect of vegetation in enhancing slope stability under seismic loading. *In Geo-Extreme 2025*. pp. 300–306. doi:[10.1061/9780784486504.030](https://doi.org/10.1061/9780784486504.030).
- Arabchobdar, F., Sadeghi, H., Gholami, M., and Alipanahi, P. 2024. Multi-directional deformation and hydraulic conductivity of expansive soils subjected to freeze–thaw cycles from three distinct initial saturation levels. *Journal of Rock Mechanics and Geotechnical Engineering*. doi:[10.1016/j.jrmge.2024.10.007](https://doi.org/10.1016/j.jrmge.2024.10.007).
- ASTM. 2009. *Annual Book of ASTM Standards*. ASTM International, West Conshohocken, Pa.
- Bodner, G., Leitner, D., and Kaul, H.P. 2014. Coarse and fine root plants affect pore size distributions differently. *Plant and Soil*, **380**: 133–151. doi:[10.1007/s11104-014-2079-8](https://doi.org/10.1007/s11104-014-2079-8). PMID: 25834289.
- Boldrin, D., Leung, A.K., and Bengough, A.G. 2017. Root biomechanical properties during establishment of woody perennials. *Ecological Engineering*, **109**: 196–206. doi:[10.1016/j.ecoleng.2017.05.002](https://doi.org/10.1016/j.ecoleng.2017.05.002).
- Bordoloi, S., Ni, J., and Ng, C.W.W. 2020. Soil desiccation cracking and its characterization in vegetated soil: a perspective review. *Science*

- of the Total Environment, **729**: 138760. doi:[10.1016/j.scitotenv.2020.138760](https://doi.org/10.1016/j.scitotenv.2020.138760). PMID: 32498161.
- Chen, R., Huang, J.W., Chen, Z.K., Xu, Y., Liu, J., and Ge, Y.H. 2019. Effect of root density of wheat and okra on hydraulic properties of an unsaturated compacted loam. *European Journal of Soil Science*, **70**(3): 493–506. doi:[10.1111/ejss.12766](https://doi.org/10.1111/ejss.12766).
- Cheng, Q., Gu, Y.D., Tang, C.S., Zhang, X.Y., and Shi, B. 2023. Desiccation cracking behavior of a vegetated soil incorporating planting density. *Canadian Geotechnical Journal*, **61**(1): 165–173. doi:[10.1139/cgj-2023-0047](https://doi.org/10.1139/cgj-2023-0047).
- Darzi, A.G., Sadeghi, H., Hedayati-Azar, A., and Alipanahi, P. 2023. Modeling water retention curve of GCL exposed to pore fluid with solute concentrations. In *International conference on environmental geotechnology, recycled waste materials and sustainable engineering*. pp. 133–144. doi:[10.1007/978-981-99-4041-7_14](https://doi.org/10.1007/978-981-99-4041-7_14).
- Dos Santos, R.A., and Esquivel, E.R. 2018. Saturated anisotropic hydraulic conductivity of a compacted lateritic soil. *Journal of Rock Mechanics and Geotechnical Engineering*, **10**(5): 986–991. doi:[10.1016/j.jrmge.2018.04.005](https://doi.org/10.1016/j.jrmge.2018.04.005).
- Fener, M., and Yesiller, N. 2013. Vertical pore structure profile of a compacted clayey soil. *Engineering Geology*, **166**: 204–215. doi:[10.1016/j.enggeo.2013.08.001](https://doi.org/10.1016/j.enggeo.2013.08.001).
- Feng, S., Liu, H.W., and Ng, C.W.W. 2019. Dimensional analysis of pore-water pressure response in a vegetated infinite slope. *Canadian Geotechnical Journal*, **56**: 1119–1133. doi:[10.1139/cgj-2018-0272](https://doi.org/10.1139/cgj-2018-0272).
- Fraccica, A., Romero, E., and Fourcaud, T. 2024. Effects of vegetation growth on soil microstructure and hydro-mechanical behaviour. *Géotechnique*, 1–41. doi:[10.1680/jgeot.23.00163](https://doi.org/10.1680/jgeot.23.00163).
- Gao, X.G., Wang, J.P., Li, X.W., Ge, S., Qiu, H.X., and Bai, M.H. 2025. Combined effects of tree canopy and root systems on soil hydraulic properties: a field monitoring of a *Populus alba*. *Rhizosphere*, **35**: 101123. doi:[10.1016/j.rhisph.2025.101123](https://doi.org/10.1016/j.rhisph.2025.101123).
- Ghestem, M., Sidle, R.C., and Stokes, A. 2011. The influence of plant root systems on subsurface flow: implications for slope stability. *Bioscience*, **61**(11): 869–879. doi:[10.1525/bio.2011.61.11.6](https://doi.org/10.1525/bio.2011.61.11.6).
- Gholami, M., Sadeghi, H., and Alipanahi, P. 2025. Anisotropic hydraulic conductivity of as-compacted, bare and vegetated soils. *Géotechnique*, **75**(6): 773–786. doi:[10.1680/jgeot.23.00248](https://doi.org/10.1680/jgeot.23.00248).
- Guo, H.W., Qu, C.X., Liu, L.L., Zhang, Q., and Liu, Y.T. 2024a. Reliability analysis of vegetated slope considering spatial variability of soil and root properties. *Computers and Geotechnics*, **169**: 106257. doi:[10.1016/j.compgeo.2024.106257](https://doi.org/10.1016/j.compgeo.2024.106257).
- Guo, H., Ng, C.W.W., and Zhang, Q. 2024b. Three-dimensional numerical analysis of plant-soil hydraulic interactions on pore water pressure of vegetated slope under different rainfall patterns. *Journal of Rock Mechanics and Geotechnical Engineering*, **16**: 3696–3706. doi:[10.1016/j.jrmge.2023.09.032](https://doi.org/10.1016/j.jrmge.2023.09.032).
- Irani, A.E., Hajjalilue-Bonab, M., Assadi-Langroudi, A., and Tabrizi, E.M. 2024. Pore-pressure-dependent performance of rocking foundations. *Soil Dynamics and Earthquake Engineering*, **183**: 108772. doi:[10.1016/j.soildyn.2024.108772](https://doi.org/10.1016/j.soildyn.2024.108772).
- Jabbarzadeh, M., Sadeghi, H., Tourchi, S., and Darzi, A.G. 2024. Thermo-hydraulic analysis of desiccation cracked soil strata considering ground temperature and moisture dynamics under the influence of soil-atmosphere interactions. *Geomechanics for Energy and the Environment*, **38**: 100558. doi:[10.1016/j.gete.2024.100558](https://doi.org/10.1016/j.gete.2024.100558).
- Jia, R., Lei, H., Hino, T., and Li, K. 2022. Changes in the permeability and permeability anisotropy of reconstituted clays under one-dimensional compression and the corresponding micromechanisms. *International Journal of Geomechanics*, **22**(2): 04021282. doi:[10.1061/\(ASCE\)GM.1943-5622.0002260](https://doi.org/10.1061/(ASCE)GM.1943-5622.0002260).
- Jotisankasa, A., and Sirirattanachai, T. 2017. Effects of grass roots on soil-water retention curve and permeability function. *Canadian Geotechnical Journal*, **54**(11): 1612–1622. doi:[10.1139/cgj-2016-0281](https://doi.org/10.1139/cgj-2016-0281).
- Karimzadeh, A.A., Leung, A.K., and Gao, Z. 2022. Shear strength anisotropy of rooted soils. *Géotechnique*, **324**: 1–14. doi:[10.1680/jgeot.22.00103](https://doi.org/10.1680/jgeot.22.00103).
- Ladd, R.S. 1978. Preparing test specimens using undercompaction. *Geotechnical Testing Journal*, **1**(1): 16–23. doi:[10.1520/GTJ10364J](https://doi.org/10.1520/GTJ10364J).
- Lann, T., Bao, H., Lan, H., Zheng, H., and Yan, C. 2024. Hydro-mechanical effects of vegetation on slope stability: a review. *Science of the Total Environment*, **926**: 171691. doi:[10.1016/j.scitotenv.2024.171691](https://doi.org/10.1016/j.scitotenv.2024.171691). PMID: 38485024.
- Leitner, D., Klepsch, S., Bodner, G., and Schnepf, A. 2010. A dynamic root system growth model based on L-Systems: tropisms and coupling to nutrient uptake from soil. *Plant and Soil*, **332**(1): 177–192. doi:[10.1007/s11104-010-0284-7](https://doi.org/10.1007/s11104-010-0284-7).
- Leung, A.K., Boldrin, D., Liang, T., Wu, Z.Y., Kamchoom, V., and Bengough, A.G. 2018. Plant age effects on soil infiltration rate during early plant establishment. *Géotechnique*, **68**(7): 646–652. doi:[10.1680/jgeot.17.T.037](https://doi.org/10.1680/jgeot.17.T.037).
- Leung, A.K., Garg, A., Coo, J.L., Ng, C.W.W., and Hau, B.C.H. 2015. Effects of roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrological Processes*, **29**(15): 3342–3354. doi:[10.1002/hyp.10452](https://doi.org/10.1002/hyp.10452).
- Li, J., Yuan, X., Ge, L., Li, Q., Li, Z., Wang, L., and Liu, Y. 2020. Rhizosphere effects promote soil aggregate stability and associated organic carbon sequestration in rocky areas of desertification. *Agriculture, Ecosystems & Environment*, **304**: 107126. doi:[10.1016/j.agee.2020.107126](https://doi.org/10.1016/j.agee.2020.107126).
- Liu, Y., Ni, J., Gu, J., Liu, S., Huang, Y., and Sadeghi, H. 2024. Influence of biopolymer-vegetation interaction on soil hydro-mechanical properties under climate change: a review. *Science of The Total Environment*, **954**: 176535. doi:[10.1016/j.scitotenv.2024.176535](https://doi.org/10.1016/j.scitotenv.2024.176535). PMID: 39332716.
- Lu, J., Zhang, Q., Werner, A.D., Li, Y., Jiang, S., and Tan, Z. 2020. Root-induced changes of soil hydraulic properties—a review. *Journal of Hydrology*, **589**: 125203. doi:[10.1016/j.jhydrol.2020.125203](https://doi.org/10.1016/j.jhydrol.2020.125203).
- Mickovski, S.B., and van Beek, L.P.H. 2009. Root morphology and effects on soil reinforcement and slope stability of young vetiver (*Vetiveria zizanioides*) plants grown in semi-arid climate. *Plant and Soil*, **324**(1–2): 43–56. doi:[10.1007/s11104-009-0130-y](https://doi.org/10.1007/s11104-009-0130-y).
- Motameni, S., Alipanahi, P., Yazdani, F., and Zhang, L. 2025. Durability assessment of biopolymer-stabilized slag tailings under freeze–thaw cycles. In *Geo-Extreme 2025*. 48–57. doi:[10.1061/9780784486535.006](https://doi.org/10.1061/9780784486535.006).
- Namadi, A.H., Hassanlourad, M., Motlagh, A.H., Alipanahi, P., Sadeghi, H., and Ahmadihosseini, A. 2020. Impact of lead and sodium carbonate on consolidation and hydraulic properties of clayey sand. In *GeoVirtual 2020. Resilience and Innovation*.
- Ng, C.W.W., Baghbanrezvan, S., Sadeghi, H., Zhou, C., and Jafarzadeh, F. 2017a. Effect of specimen preparation techniques on dynamic properties of unsaturated fine-grained soil at high suctions. *Canadian Geotechnical Journal*, **54**(9): 1310–1319. doi:[10.1139/cgj-2016-0531](https://doi.org/10.1139/cgj-2016-0531).
- Ng, C.W.W., Kamchoom, V., and Leung, A.K. 2016a. Centrifuge modelling of the effects of root geometry on transpiration-induced suction and stability of vegetated slopes. *Landslides*, **13**: 925–938. doi:[10.1007/s10346-015-0645-7](https://doi.org/10.1007/s10346-015-0645-7).
- Ng, C.W.W., Leung, A.K., and Woon, K.X. 2014. Effects of soil density on grass-induced suction distributions in compacted soil subjected to rainfall. *Canadian Geotechnical Journal*, **51**(3): 311–321. doi:[10.1139/cgj-2013-0221](https://doi.org/10.1139/cgj-2013-0221).
- Ng, C.W.W., Liu, H.W., and Feng, S. 2015. Analytical solutions for calculating pore-water pressure in an infinite unsaturated slope with different root architectures. *Canadian Geotechnical Journal*, **52**: 1981–1992. doi:[10.1139/cgj-2015-0001](https://doi.org/10.1139/cgj-2015-0001).
- Ng, C.W.W., Ni, J.J., and Leung, A.K. 2020a. Effects of plant growth and spacing on soil hydrological changes: a field study. *Géotechnique*, **70**(10): 867–881. doi:[10.1680/jgeot.18.P.207](https://doi.org/10.1680/jgeot.18.P.207).
- Ng, C.W.W., Ni, J.J., Leung, A.K., and Wang, Z.J. 2016b. A new and simple water retention model for root-permeated soils. *Géotechnique Letters*, **6**(1): 106–111. doi:[10.1680/jgele.15.00187](https://doi.org/10.1680/jgele.15.00187).
- Ng, C.W.W., Ni, J.J., Leung, A.K., Zhou, C., and Wang, Z.J. 2017b. Effects of planting density on tree growth and induced suction. *Géotechnique*, **66**(9): 711–724. doi:[10.1680/jgeot.15.P.196](https://doi.org/10.1680/jgeot.15.P.196).
- Ng, C.W.W., Sadeghi, H., Jafarzadeh, F., Sadeghi, M., Zhou, C., and Baghbanrezvan, S. 2020b. Effect of microstructure on shear strength and dilatancy of unsaturated loess at high suctions. *Canadian Geotechnical Journal*, **57**(2): 221–235. doi:[10.1139/cgj-2018-0592](https://doi.org/10.1139/cgj-2018-0592).
- Ng, C.W.W., Zhang, Q., Zhang, S., Lau, S.Y., Guo, H., and Li, Z. 2024. A new state-dependent constitutive model for cyclic thermo-mechanical behaviour of unsaturated vegetated soil. *Canadian Geotechnical Journal*, **61**: 2155–2179. doi:[10.1139/cgj-2023-0268](https://doi.org/10.1139/cgj-2023-0268).
- Ni, J.J., Leung, A.K., and Ng, C.W.W. 2019. Influences of plant spacing on root tensile strength of *Schefflera arboricola* and soil shear

- strength. *Landscape and Ecological Engineering*, **15**: 223–230. doi:[10.1007/s11355-019-00374-x](https://doi.org/10.1007/s11355-019-00374-x).
- Ni, J.J., Leung, A.K., Ng, C.W.W., and So, P.S. 2017. Investigation of plant growth and transpiration-induced matric suction under mixed grass-tree conditions. *Canadian Geotechnical Journal*, **54**(4): 561–573. doi:[10.1139/cgj-2016-0226](https://doi.org/10.1139/cgj-2016-0226).
- Önal, Y., Kayadelen, C., Serin, S., and Altay, G. 2025. Road sub-grade stabilization with cement-coated geotextile under static and cyclic loading. *Geosynthetics International*. doi:[10.1680/jgein.25.00024](https://doi.org/10.1680/jgein.25.00024).
- Ozturk, E., Turkezer, M., Duman, E., Saricicek, Y.E., Kaya, H.D., Umut Arica, E., and Pekcan, O. 2022. Visualization of Atterberg limits test through a virtual soil laboratory. In *Geo-Congress 2022*. pp. 505–514. doi:[10.1061/9780784484067.050](https://doi.org/10.1061/9780784484067.050).
- Qu, C., Ng, C.W.W., and Guo, H. 2025. Reliability-based multi-objective optimisation for the design of vegetation planting strategy on landfill cover systems. *Computers and Geotechnics*, **187**: 107483. doi:[10.1016/j.compgeo.2025.107483](https://doi.org/10.1016/j.compgeo.2025.107483).
- Rahardjo, H., Satyanaga, A., Leong, E.C., Santoso, V.A., and Ng, Y.S. 2014. Performance of an instrumented slope covered with shrubs and deep-rooted grass. *Soils and Foundations*, **54**(3): 417–425. doi:[10.1016/j.sandf.2014.04.010](https://doi.org/10.1016/j.sandf.2014.04.010).
- Sadeghi, H., and Alipanahi, P. 2020. Saturated hydraulic conductivity of problematic soils measured by a newly developed low-compliance tri-axial permeameter. *Engineering Geology*, **278**: 105827. doi:[10.1016/j.enggeo.2020.105827](https://doi.org/10.1016/j.enggeo.2020.105827).
- Sadeghi, H., Gholami, M., Alipanahi, P., and Song, D. 2024a. The influence of isotropic loading and unloading on anisotropic evolution of saturated hydraulic conductivity of bentonite-sand mixtures in a cube triaxial permeameter. *Engineering Geology*, **331**, 107454. doi:[10.1016/j.enggeo.2024.107454](https://doi.org/10.1016/j.enggeo.2024.107454).
- Sadeghi, H., Kohal, F.Y.B., Gholami, M., Alipanahi, P., and Song, D. 2023. Hydro-mechanical modeling of a vegetated slope subjected to rainfall. *E3S Web of Conferences*, **382**: 13004. doi:[10.1051/e3sconf/202338213004](https://doi.org/10.1051/e3sconf/202338213004).
- Sadeghi, H., Motlagh, A.H., Darzi, A.G., Alipanahi, P., Khodadadi, S., and Song, D. 2024b. Measurement and modeling of the saturated hydraulic conductivity functions of collapsible and expansive soils. *International Journal of Geomechanics*, **24**(4): 04024039. doi:[10.1061/IJGNALGMENG-8788](https://doi.org/10.1061/IJGNALGMENG-8788).
- Scholl, P., Leitner, D., Kammerer, G., Lioskandl, W., Kaul, H.P., and Bodner, G. 2014. Root induced changes of effective 1D hydraulic properties in a soil column. *Plant and Soil*, **381**(1–2): 193–213. doi:[10.1007/s11104-014-2121-x](https://doi.org/10.1007/s11104-014-2121-x). PMID: 25834290.
- Shafiee, A. 2008. Permeability of compacted granule-clay mixtures. *Engineering Geology*, **97**(3–4): 199–208. doi:[10.1016/j.enggeo.2008.01.002](https://doi.org/10.1016/j.enggeo.2008.01.002).
- Shao, W., Ni, J., Leung, A.K., Su, Y., and Ng, C.W.W. 2017. Analysis of plant root-induced preferential flow and pore-water pressure variation by a dual-permeability model. *Canadian Geotechnical Journal*, **54**(11): 1537–1552. doi:[10.1139/cgj-2016-0629](https://doi.org/10.1139/cgj-2016-0629).
- Sitarenios, P., Casini, F., Askarinejad, A., and Springman, S. 2021. Hydro-mechanical analysis of a surficial landslide triggered by artificial rainfall: the Ruedlingen field experiment. *Géotechnique*, **71**(2): 96–109. doi:[10.1680/jgeot.18.P.188](https://doi.org/10.1680/jgeot.18.P.188).
- Song, L., Li, J.H., Zhou, T., and Fredlund, D.G. 2017. Experimental study on unsaturated hydraulic properties of vegetated soil. *Ecological Engineering*, **103**: 207–216. doi:[10.1016/j.ecoleng.2017.04.013](https://doi.org/10.1016/j.ecoleng.2017.04.013).
- Tsiampousi, A. 2024. A three-dimensional study of vegetation management on cut slopes. *Canadian Geotechnical Journal*, **62**: 1–16. doi:[10.1139/cgj-2023-0220](https://doi.org/10.1139/cgj-2023-0220).
- Wang, J.P., Sha, J.F., Gao, X.G., Dadda, A., Qi, M., and Ge, S. 2024. Experimental investigation of the microscopic interaction mechanism between sand and tall fescue roots by using X-ray microtomography. *Acta Geotechnica*, **19**(3): 1529–1554. doi:[10.1007/s11440-024-02234-x](https://doi.org/10.1007/s11440-024-02234-x).
- Wang, X., Ma, C., Wang, Y., Wang, Y., Li, T., Dai, Z., and Li, M. 2020. Effect of root architecture on rainfall threshold for slope stability: variabilities in saturated hydraulic conductivity and strength of root-soil composite. *Landslides*, **17**: 1965–1977. doi:[10.1007/s10346-020-01422-6](https://doi.org/10.1007/s10346-020-01422-6).
- Wu, Z., Leung, A.K., Boldrin, D., and Ganesan, S.P. 2021. Variability in root biomechanics of *Chrysopogon zizanioides* for soil eco-engineering solutions. *Science of the Total Environment*, **776**: 145943. doi:[10.1016/j.scitotenv.2021.145943](https://doi.org/10.1016/j.scitotenv.2021.145943).
- Xiao, T., Li, P., Fei, W., and Wang, J. 2023. Effects of vegetation roots on the structure and hydraulic properties of soils: a perspective review. *Science of the Total Environment*, 167524. doi:[10.1016/j.scitotenv.2023.167524](https://doi.org/10.1016/j.scitotenv.2023.167524). PMID: 37793452.
- Yan, W.M., and Zhang, G. 2015. Soil-water characteristics of compacted sandy and cemented soils with and without vegetation. *Canadian Geotechnical Journal*, **52**(9): 1331–1344. doi:[10.1139/cgj-2014-0334](https://doi.org/10.1139/cgj-2014-0334).
- Yazdani, F., Alipanahi, P., and Sadeghi, H. 2024a. A comparative study of environmental and economic assessment of vegetation-based slope stabilization with conventional methods. *Journal of Environmental Management*, **359**: 121002. doi:[10.1016/j.jenvman.2024.121002](https://doi.org/10.1016/j.jenvman.2024.121002). PMID: 38696847.
- Yazdani, F., Sadeghi, H., Alipanahi, P., Gholami, M., and Leung, A.K. 2024b. Evaluation of plant growth and spacing effects on bioengineered slopes subjected to rainfall. *Biogeotechnics*, **2**: 100080. doi:[10.1016/j.bgtech.2024.100080](https://doi.org/10.1016/j.bgtech.2024.100080).
- Yildiz, A., Graf, F., and Springman, S.M. 2020. On the dilatancy of root-permeated soils under partially saturated conditions. *Géotechnique Letters*, **10**(2): 227–230. doi:[10.1680/jgele.19.00048](https://doi.org/10.1680/jgele.19.00048).
- Zarei, A., Hasani Motlagh, A., Hajialilue Bonab, M., and Alipanahi, P. 2024. Influence of ground slope and relative density on liquefaction-induced lateral spreading: a physical modeling. *Geotechnical and Geological Engineering*, **42**(6): 5189–5208. doi:[10.1007/s10706-024-02836-y](https://doi.org/10.1007/s10706-024-02836-y).
- Zhu, J., Leung, A.K., and Wang, Y. 2023. Partially drained uplift behaviour of plant roots in dilative soils. *Canadian Geotechnical Journal*, **61**(3): 500–518. doi:[10.1139/cgj-2023-0104](https://doi.org/10.1139/cgj-2023-0104).